Fiscal Year:	FY 2004	**Task Last Updated:**	FY 10/24/2006
PI Name:	Peterson, G.P. Ph.D.		
Project Title:	An Experimental and Analytical Investigation of Condensing Radiators for Application to Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	09/01/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sobhan, Choondal (National Institute of Technology Calicut)		
Grant/Contract No.:	NNC05GA28G		
Performance Goal No.:			
Performance Goal Text:			
	Because of the low weight to power dissipation ratio and the high degree of temperature uniformity that can be obtained, condensing radiators have numerous space based applications. In these applications, condensation heat transfer could be used to transfer heat from the source to the radiators, where through phase change, the desired heat transfer can be achieved and the waste heat ultimately dissipated from the radiator surface to space. While the ultimate heat dissipation from the radiator is by thermal radiation, the use of phase change and in particular, the condensation process that occurs in the condenser tubes serves to spread the heat uniformly over the radiator surface. Theoretical analysis of such a system must therefore consider the phase change phenomenon, the condensate film formation and the two- phase flow inside the tubes, in addition to the actual radiative heat transfer to the surroundings. Such an analysis, validated against appropriate experimental data, is essential to the optimization of condensing radiator design and can serve as the		

Task Description:	measure by which these phase change radiators can be compared to pumped, single phase radiators. The current proposal focuses on an extensive theoretical study of the condensing radiators, and validation against the experimental data. Based on the computational results, it is proposed to develop not only a more clear understanding of the behavior occurring in these phase change tubes, but also to develop an optimum design with respect to the geometric and operational parameters, and the level of microgravity under which the system is designed to operate. The combination of analytical and experimental studies proposed in this investigation will yield an analytical model, verified by quantitative experimental results, which can be used in the design of condensing radiators for microgravity applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004. Added to Task Book in 2006 when received information on the project.
Bibliography Type:	Description: (Last Updated:)